INDEX

Student : V. Bhavya Class : ..... Sub : ..... Roll No : .....

S.No.	Date	TITLE	Page No.	Teacher's Sign/ Remarks
1	20-1-26	R Variables		
2	20-1-26	R Data types		
3	20-1-26	R Numbers		
4	20-1-26	R Math		
5	27-1-26	R Strings		
6	27-1-26	R Booleans		
7	27-1-26	R Operators		
8	18-2-26	Basics of R		
9	18-2-26	Operators & Decision Making		
10	24-2-26	Functions		
11	10-3-26	Data frames		
12	10-3-26	String Manipulation		
13	17-3-26	Data Structures		
14	24-3-26	Data in a file		
15	24-3-26	Pie charts and bar graphs		
16	24-3-26	Statistical Analysis		

R PROGRAMMING LAB OUTPUTS

EXERCISE-1

1.
```
[1] 205
[1] "Bhavya"
[1] 9.12
[1] TRUE
```

2.
```
[1] 6205
[1] 52500.8
[1] "HR"
[1] TRUE
```

3.
```
[1] "Smartphone"
[1] 28999
[1] 18
[1] TRUE
```

4.
```
[1] "Hyderabad"
[1] 32.7
[1] TRUE
[1] 15.4
```

5.
```
[1] "Physics"
[1] 100
[1] 86
[1] TRUE
```

6.
```
[1] 987654321
[1] "Bhavya"
[1] 35750.75
[1] TRUE
```

7.
```
[1] "Bahubali"
[1] 9.1
[1] 1850
[1] TRUE
```

8.
```
[1] 420
[1] "City Center"
[1] 38.5
[1] TRUE
```

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EXERCISE-2

1. `[1] "Eligible for Loan"`

2. `[1] "Exam Allowed"`

3. `[1] "Discount Applied"`

4. `[1] "High Usage"`

5. `[1] "Bonus Eligible"`

6. `[1] "Valid Password"`

7. `[1] "Extreme Temperature Warning"`

8. `[1] "A"`

9. `[1] "Credit Card Approved"`

10. `[1] "Medium Risk"`

11. `[1] "Long Green Signal"`

12. `[1] "Eligible for Appraisal"`

13. `[1] "Exam Allowed"`

14. `[1] "Surge Pricing Applied"`

15. `[1] "Machine Shutdown"`

16. `[1] "Admission Granted"`

EXERCISE-3

1. `[1] 2700`

2. `[1] "Pass"`

3. `[1] 37800`

4. `[1] 875`

5. `[1] 4960`

6. `[1] 82.4`

7. `[1] "B"`

8. `[1] "Eligible to Vote"`

9. `[1] 8118.39`

10. `[1] 11235`

11. `[1] 5040`

12. `[1] "Distinction"`

13. `[1] 1782`

14. `[1] 9360`

15. `[1] 54912`

16. `[1] "High Usage"`

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EXERCISE-4

1.

	roll_no	name	dept	Marks
1	101	Bhavya	IT	88
2	102	Anil	CSE	76
3	103	Priya	ECE	91

	roll_no	name	dept	Marks
1	101	Bhavya	IT	88
2	102	Anil	CSE	76
3	103	Priya	ECE	91
4	104	Rahul	IT	85

2

	emp_id	name	dept	Salary
1	1	Bhavya	HR	35000
2	2	Sneha	IT	45000
3	3	Raj	Finance	50000

	emp_id	name	dept	Salary
1	1	Bhavya	HR	35000
2	2	Sneha	IT	45000
3	3	Raj	Finance	50000
4	4	Anil	IT	48000

3

	patient_id	name	age	Cost
1	201	Bhavya	20	12000
2	202	Lakshmi	38	15000

	patient_id	name	age	Cost
1	201	Bhavya	20	12000
2	202	Lakshmi	38	15000
3	203	Arjun	25	18000

4

	product_id	product_name	Quantity
1	1	Pen	100
2	2	Book	50
3	3	Bag	30

	product_id	product_name	Quantity
1	1	Pen	100
2	2	Book	50
3	3	Bag	30
4	4	Bottle	75

5

	roll_no	name	dept	Sem1	Sem2	Attendance
1	1	Bhavya	IT	78	80	92
2	2	Bina	ECE	82	85	88
3	3	Charan	CSE	90	88	95

	roll_no	name	dept	Sem1	Sem2	Attendance
1	1	Bhavya	IT	78	80	92
2	2	Bina	ECE	82	85	88
3	3	Charan	CSE	90	88	95
4	4	Anil	IT	85	87	90

6

	emp_id	name	dept	Basic	HRA	DA	Rating
1	101	Bhavya	IT	30000	6000	3000	4.5
2	102	Sneha	HR	35000	7000	3500	4.2

	emp_id	name	dept	Basic	HRA	DA	Rating
1	101	Bhavya	IT	30000	6000	3000	4.5
2	102	Sneha	HR	35000	7000	3500	4.2
3	103	Kiran	Finance	40000	8000	4000	4.8

7

	patient_id	name	age	Disease	Amount
1	201	Bhavya	20	Diabetes	15000
2	202	Lakshmi	38	Hypertension	12000

	patient_id	name	age	Disease	Amount
1	201	Bhavya	20	Diabetes	15000
2	202	Lakshmi	38	Hypertension	12000
3	203	Arjun	25	Cardiac	25000

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8

```
  product_id product_name UnitsSold Revenue
1          1       Mobile       120 1200000
2          2       Laptop        80 4000000
3          3       Tablet        60 1800000
  product_id product_name UnitsSold Revenue
1          1       Mobile       120 1200000
2          2       Laptop        80 4000000
3          3       Tablet        60 1800000
4          4  Smart Watch        90  900000
```

9

```
  field_id  crop Moisture Temp ExpectedYield
1        1  Rice       45   32           4.5
2        2 Wheat       38   29           3.8
  field_id  crop Moisture Temp ExpectedYield
1        1  Rice       45   32           4.5
2        2 Wheat       38   29           3.8
3        3 Maize       40   31           4.1
```

10

```
  RollNo   Name Dept Sem1 Sem2 Attendance
1      1 Bhavya   IT   78   80         92
2      2   Bina  ECE   82   85         88
3      3 Charan  CSE   90   88         95
```

EXERCISE-5

1

```
[1] "High Usage"
```

2

```
[1] "bhavya" "anil"   "priya"
```

3

```
[1] "Bhavya"       "Data Science" "AI"
```

4

```
[1]  TRUE  TRUE FALSE
```

5

```
[1] "Strong Password"
```

6

```
[1] 5
```

7

```
[1] "csv" "pdf" "png"
```

8

```
[1] "Bha" "Kar" "Ani"
```

9

```
[1] "Bhavya" "Bala"   "Ravi"
```

10

```
    Name           Email Department Username ValidEmail
1 BHAVYA bhavya@univ.edu         IT   bhavya       TRUE
2   RAVI   ravi@univ.edu        ECE     ravi       TRUE
```

11

```
      Product        Code ProductCode
1 Apple Phone PRD-101-APL         APL
2  Samsung-Tv PRD-202-SAM         SAM
3 Dell Laptop PRD-303-DEL         DEL
```

12

```
  User             Comment                 CleanComment
1    A     this app is bad     this app is unacceptable
2    B very bad experience very unacceptable experience
3    C      not bad at all      not unacceptable at all
```

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EXERCISE-6

1

```
[1] 434
[1] 86.8
[1] 92
```

2

```
[1] 31
[1] 36
```

3

```
$ID
[1] 101

$Name
[1] "Bhavya"

$Salary
[1] 45000

$Department
[1] "IT"

[1] "Bhavya"
[1] 45000
```

4

```
$PatientID
[1] 202

$Name
[1] "Bhavya"

$Age
[1] 20

$TestResults
[1] 120 140 130

[1] 120 140 130
```

5

```
  RollNo    Name Dept Marks
1      1  Bhavya   IT    85
2      2 Karthik  ECE    78
3      3   Meena  CSE    92
```

6

```
[1] 170000
```

7

```
[1] 1200  850  560
$CustomerID
[1] 301

$Name
[1] "Bhavya"

$City
[1] "Chennai"

     Item Price
1   Phone  1200
2 Headset   850
3 Charger   560
```

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8

```
$Name
[1] "Bhavya"

$RollNo
[1] 105

$Marks
[1] 78 85 88

  RollNo Total
1    101   240
2    102   255
3    103   270
```

9

```
[1] 5000 5200 5100
$AccountNo
[1] 12345

$Name
[1] "Bhavya"

   Date Amount
1 01-01   -500
2 02-01   1000
3 03-01   -200
```

10

```
[1] 85 90 88
$Name
[1] "Bhavya"

$Age
[1] 20

$Scores
[1] 85 90 88

  Subject Marks
1    Math    85
2      CS    90
3 Physics    88
```

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EXERCISE-7

Step 1: Initial Data View

Date	Product	Category	Quantity	Price	TotalSales
2026-01-01	Phone	Electronics	5	20000	100000
2026-01-01	Bag	Accessories	10	1500	15000
2026-01-02	Laptop	Electronics	3	50000	150000
2026-01-02	Headset	Electronics	7	2500	17500
2026-01-03	Bag	Accessories	5	1500	7500
2026-01-03	Phone	Electronics	3	20000	60000

Step 2: Cleaned Data with TotalSales

Date	Product	Category	Quantity	Price	TotalSales
2026-01-01	Phone	Electronics	5	20000	100000
2026-01-01	Bag	Accessories	10	1500	15000
2026-01-02	Laptop	Electronics	3	50000	150000
2026-01-02	Headset	Electronics	7	2500	17500
2026-01-03	Bag	Accessories	5	1500	7500
2026-01-03	Phone	Electronics	3	20000	60000

Step 3: Total Sales per Product

Product	TotalSales
Bag	22500
Headset	17500
Laptop	150000
Phone	160000

Step 4: Top-Selling Product

Product	TotalSales
Phone	160000

Step 5: Quantity per Category

Category	Quantity
Accessories	15
Electronics	18

Step 6: High-Value Products

Product	TotalSales
Bag	22500
Headset	17500
Laptop	150000
Phone	160000

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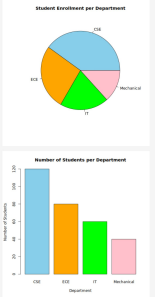
EXERCISE-8

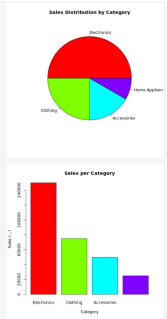
1

Output

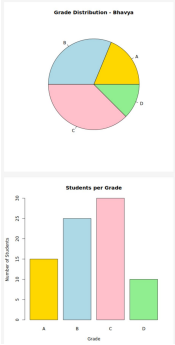
```
[Execution complete with exit code 0]
```



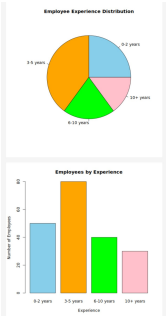
2



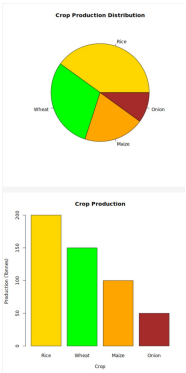
3



4



5



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EXERCISE-9

1

```
 Maths Science English
  82.9    80.0    85.4
 Maths Science English
  85.5    81.0    86.0
  Maths  Science  English
9.302927 9.809293 7.183314
   Name Maths Science English Total
4 Anitha    95      92      98   285
      Name Maths Science English Total
1    Bhavya    85      78      92   255
2     Meena    90      88      80   258
4    Anitha    95      92      98   285
6     Sneha    88      80      85   253
8  Gayathri    92      90      91   273
10  Lakshmi    86      82      87   255
```

2

```
   Product Mean Max Min Variance
1    Phone  5.4   7   4      1.3
2   Laptop  3.4   5   2      1.3
3  Headset  6.6   8   5      1.3
4      Bag 10.0  12   8      2.5
5   Camera  2.2   3   1      0.7
6    Mouse  7.6   9   6      1.3
7 Keyboard  6.4   8   5      1.3
  Product Mean Max Min Variance
4     Bag   10  12   8      2.5
```

3

```
[1] 45.4
[1] 46.5
[1] 29 60
[1] 9.547542
  PatientID   Name Age
3       203 Bhavya  52
5       205   Ravi  60
7       207  Kiran  55
```

4

```
     Dept Salary.Mean Salary.Median
1 Finance    42333.33         42000
2      HR    36000.00         36000
3      IT    31500.00         31500
  EmpID    Name    Dept Salary
3   103   Kiran Finance  40000
6   106  Bhavya Finance  42000
8   108 Lakshmi      HR  37000
9   109   Arjun Finance  45000
```

5

```
[1] 4.1
[1] 4
[1] 4
[1] 0.5444444
  ProductID Product Rating
2         2  Laptop      5
5         5  Camera      5
8         8 Charger      5
```

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